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To cite this article: Vladimir N. Anisimov, Lev M. Berstein, Peter A. Egormin, Tatiana S. Piskunova, Irina G. Popovich, Mark A. Zabezhinski, Margarita L. Tyndyk, Maria V. Yurova, Irina G. Kovalenko, Tatiana E. Poroshina & Anna V. Semenchenko (2008) Metformin slows down aging and extends life span of female SHR mice, *Cell Cycle*, 7:17, 2769-2773, DOI: [10.4161/cc.7.17.6625](https://doi.org/10.4161/cc.7.17.6625)

To link to this article: <https://doi.org/10.4161/cc.7.17.6625>



Published online: 01 Sep 2008.



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Report

Metformin slows down aging and extends life span of female SHR mice

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Key words: metformin, biomarkers of aging, life extension, spontaneous carcinogenesis, mice

Studies in mammals have led to the suggestion that hyperglycemia and hyperinsulinemia are important factors both in aging and in the development of cancer. It is possible that the life-prolonging effects of calorie restriction are due to decreasing IGF-1 levels. A search of pharmacological modulators of insulin/IGF-1 signaling pathway (which resemble effects of life span extending mutations or calorie restriction) could be a perspective direction in regulation of longevity. Antidiabetic biguanides are most promising among them. Here we show the chronic treatment of female outbred SHR mice with metformin (100 mg/kg in drinking water) slightly modified the food consumption but decreased the body weight after the age of 20 months, slowed down the age-related switch-off of estrous function, increased mean life span by 37.8%, mean life span of last 10% survivors by 20.8%, and maximum life span by 2.8 months (+10.3%) in comparison with control mice. On the other side, treatment with metformin failed influence blood estradiol concentration and spontaneous tumor incidence in female SHR mice. Thus, antidiabetic biguanide metformin dramatically extends life span, even without cancer prevention in this model.

Introduction

The potential link between aging and insulin/IGF-1 signaling has attracted substantial attention during last years. The potential connection was evidenced by an increase in incidence of insulin resistance and type 2 diabetes in accelerated aging syndromes, on the one side, as well as by life span extension due to caloric restriction (CR) in rodents, on the other. Concomitant reduction in plasma insulin and plasma glucose levels, which implies increased sensitivity to insulin, emerges as a hallmark of increased longevity.¹ Hyperglycemia is an important aging factor involved in generation of advanced glycosylation endproducts (AGEs).^{2,3} There is evidence that hyperinsulinemia favors accumulation of oxidized protein by reducing its degradation as well as facilitates protein oxidation by increasing steady-state level of oxidative stress.² Untreated diabetics with elevated glucose levels suffer many manifestations of accelerated

aging, such as impaired wound healing, obesity, cataracts, vascular and microvascular damage.⁴ It is important to stress that hyperinsulinemia is an significant factor not only in aging but also in the development of cancer.⁴⁻⁷

The concept of CR mimetics is now being intensively explored.⁸⁻¹⁰ CR mimetics involve interventions that produce physiological and anti-aging effects similar to CR. It was suggested to use biguanide antidiabetics as a potential anti-aging treatment.^{4,11-15} The antidiabetic drugs, phenformin and buformin, were observed to reduce hyperglycemia and produce the following effects: improved glucose utilization; reduced free fatty acid utilization, gluconeogenesis, serum lipids, insulin and IGF-1, and reduced body weight both in humans and experimental animals.^{4,16} The use of phenformin (phenethylbiguanide) in humans has been limited during the last two-three decades because of a potential association with lactic acidosis. Metformin (dimethylbiguanide) does not increase risk for lactic acidosis or for increased lactate levels in type 2 diabetes,¹⁷ but may have some adverse effects, including renal insufficiency¹⁸ and gastrointestinal disturbances.¹⁹ In transgenic HER-2/neu mice, it was shown that this widely used antidiabetic biguanide slowed down aging and tumor development.²⁰ Treatment with metformin prolongs life span of rats (G.Roth, personal communication) and mice.^{21,22}

In this paper we present results of experiments with the antidiabetic biguanide metformin on some ageing related biological parameters, survival and spontaneous tumorigenesis in female outbred SHR mice.

Results

Age-related body weight dynamics. The body weight of mice in both control and metformin-treated groups increased with age, exceeding by 12 months the body weight of 3-month-old animals by 25.1% in the control group, and by 35.1% in the group treated with metformin. There was no difference in the mean body weight of mice exposed and non-exposed to the drug until the age of 20 months and a tendency to a decrease of the body weight was observed in metformin treated group after this age (Fig. 1).

Age-related dynamics of food and water consumption. The amount of food daily consumed by mice during the period of observation was similar in the control group and in metformin-treated group until the age of 1 year. However, at the age of 12 to 16 months the food consumption was increased by 14–22% ($p < 0.05$) and at the age of 22 months it was decreased by 55% in metformin-treated group as compared with the control (Fig. 2). The amount of water

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Submitted: 06/05/08; Accepted: 06/11/08

Previously published online as a *Cell Cycle* E-publication:
<http://www.landesbioscience.com/journals/cc/article/6625>

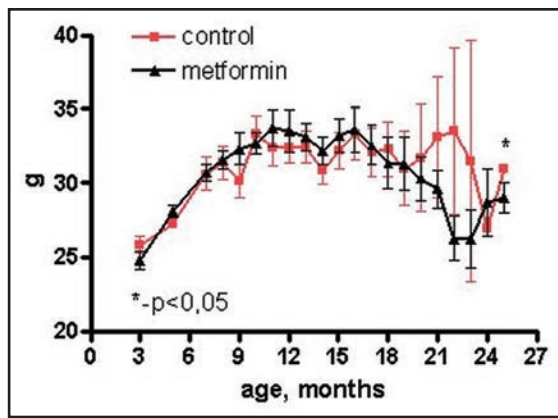


Figure 1. Age-related dynamics of body weight (g) in SHR mice treated and not treated with metformin.

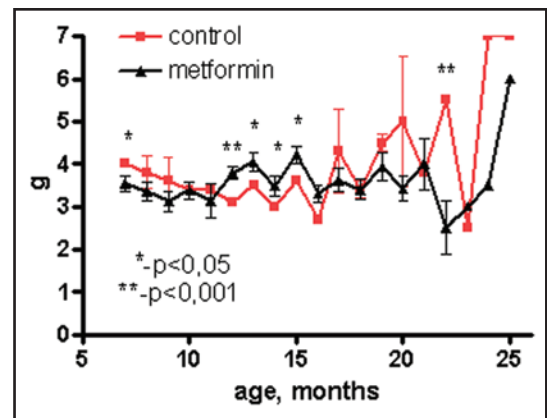


Figure 2. Age-related dynamics of food consumption in female SHR mice treated or non-treated with metformin.

consumed by mice varied during the period of observation and was practically same in both groups (data are not shown).

Age-related dynamics of estrous function in mice. The length of estrous cycles in the control female SHR mice was not significantly changed with age, whereas it was increased at the age of 18 months of age in mice exposed to metformin (Table 1). The fraction of mice with regular estrous cycles decreased with the age in the control mice but not in mice treated with metformin (Table 1).

Age-related dynamics of body temperature in mice. There was no difference in average body temperature between control and metformin-treated groups during the entire period of observation (data not shown).

Effect of metformin on metabolic and hormonal parameters in mice. Between the age of 3 and 12 months the average value of serum level of glucose and triglycerides was not changed, whereas the level of total cholesterol decreased by 31.5%, $p < 0.05$ and the level of insulin was increased in the control group (Table 2). The treatment with metformin slightly decreased the level of triglycerides ($p < 0.05$) but failed to influence other metabolic parameters. The level of estradiol-17 β was similar in 3- and 9-month-old control mice 5.96 ± 0.76 pg/ml and 6.86 ± 1.44 pg/ml, correspondingly, and it was 8.03 ± 1.46 pg/ml in 9-month-old mice treated with metformin ($p > 0.05$).

Survival and longevity of female SHR mice. According to the log-rank test³⁰ the difference in survival of female SHR mice subjected to metformin treatment, compared to the control group, is significant (p -value is 0.000744). Survival dynamics in control and metformin-treated mice are shown in Figure 3. Only one control mouse (2%) survived the age of 24 months whereas 10 mice (20%) survived this age in metformin-treated group ($p < 0.001$).

Metformin treatment significantly increased mean ($+37.9\%$, $p < 0.01$) and maximum life span ($+10.3\%$) of female SHR mice (Table 3). Mean life span of long-living individuals (last 10% of survivors) was significantly greater in the group exposed to metformin ($+20.8\%$) compared to the control one. Parameter α of the Gompertz model, which is interpreted as the rate of aging, was higher (by 1.7 times) in the group subjected to metformin treatment than in controls, whereas the mortality rate doubling time (MRDT) was decreased in metformin-treated group of mice. The difference between the control and the experimental group was statistically significant

Table 1 **Effect of metformin on age-related dynamics of estrus functional parameters in SHR mice**

Age, months	No. of mice	Length of estrous cycle (days)	Rate of estrous cycles of various length (%)			Fraction of mice with regular cycles (%)
			<5 days	5–7 days	>7 days	
Control						
3	50	6.9 ± 0.47	12	47	41	74
6	39	6.7 ± 0.28	6	63	31	80
9	32	6.3 ± 0.27	23	50	27	81
12	21	6.4 ± 0.31	5	74	21	71
15	17	7.5 ± 0.67	9	55	36	59
18	10	6.3 ± 0.48	0	67	33	50
21	3	-	-	-	-	33
Metformin						
3	50	6.0 ± 0.31	20	63	17	88
6	50	6.3 ± 0.28	18	57	25	88
9	45	6.2 ± 0.22	12	66	22	78
12	32	6.4 ± 0.24	12	69	19	97*
15	26	6.4 ± 0.30	20	45	35	77
18	20	7.6 ± 0.39*	0	71	29	75*
21	8	7.1 ± 0.70	0	71	29	88*

The difference with the controls of corresponding age in the control group: * < 0.05 (chi-square test).

(Table 3). Parameters for the Cox's proportional hazard model in the context of "control & metformin" was estimated as following: $\beta = -0.701$; $\text{Exp}(\beta) = 0.496$; $\text{se}(\beta) = 0.212$, p -value = 9.40×10^{-4} . Thus, relative risk of death decreased significantly on the treatment with metformin.

Spontaneous tumor development in mice SHR mice. According to the long-rank test³⁰ there were no significant differences in age-related distributions of the total tumors occurrence in control and metformin-treated groups (Fig. 4). The first tumor-bearing mice died at the age of 316 days in the metformin-treated group and at the 457 days in the control group. Total tumor incidence in effective control

Table 2 Effect of metformin on metabolic parameters in the serum of female SHR mice*

Treatment	Age, months	Glucose mM/l	Total cholesterol mM/l	Triglycerides mM/l	Insulin Mku/ml
Control	3	6.05 ± 0.44	3.21 ± 0.27	1.05 ± 0.08	1.00 ± 0.83
	12	6.23 ± 0.57	2.20 ± 0.15 ^a	1.13 ± 0.06	10.27 ± 3.41
Metformin	12	5.83 ± 0.76	2.43 ± 0.20	0.94 ± 0.04 ^{ab}	13.28 ± 5.41

There were 10 animals in each group. ^aThe difference with the controls at the age of 3 months is significant (Student's *t* tests): *p* < 0.05. ^bThe difference with the controls at the age of 12 months is significant (Student's *t* tests): *p* < 0.05.

Table 3 Parameters of life span in female SHR mice treated and not treated with metformin

Parameters	Control	Metformin
Number of mice	50	50
Mean life span, days (M ± S.E.)	388 ± 29.2	535 ± 31.9 (+37.9%)
Median	297	570 (+91.9%)
Mean life span of last 10% of survivors, days	727 ± 22.5	878 ± 6.6 (+20.8%)
Maximum life span, days	814	898 (+10.3%)
$\alpha \times 10^3$, days ⁻¹	2.4439 (2.0331; 2.26013)	4.1563 (4.0056; 4.3957)*
MRDT, days	283.6 (266.5; 340.9)	166.8 (157.7; 173.0)*

*The differences with the controls are significant, *p* < 0.01 (Student *t* test).

female mice (survival by the time of the death from the first tumor in the experiment) was 65.2% and respectively 55% in metformin-treated animals (Table 4). Mammary carcinomas and leukemias developed most frequently, in accord with oncological characteristics of the female SHR mice.³¹ Treatment with metformin failed to influence the total incidence of malignant tumors (Table 4). The mean latent period of all or only malignant tumors was similar in both groups. There was no significant difference in the incidence of any other tumors between mice treated with metformin and controls.

According to the long-rank test the difference in survival of the tumor-free SHR mice subjected to the metformin treatment, compared to the tumor-free mice from the control group is significant (*p*-value = 0.000216); the mean life span was 290 ± 26.2 days in tumor-free control mice and 494 ± 50.6 days in metformin-treated group (+70.3%, *p* < 0.001).

Discussion

Our experiments have shown that long-term treatment with the antidiabetic biguanide metformin significantly increased mean life span of all as well as 10% most long-living female SHR mice. These observations are in agreement with the data previously obtained with metformin in relatively short-living transgenic HER-2/neu mice²⁰ and with another antidiabetic biguanides, phenformin and buformin. It was shown in earlier studies that phenformin and buformin increase the life span in female C3H/Sn mice¹¹ and in female outbred rats.^{12,13,31}

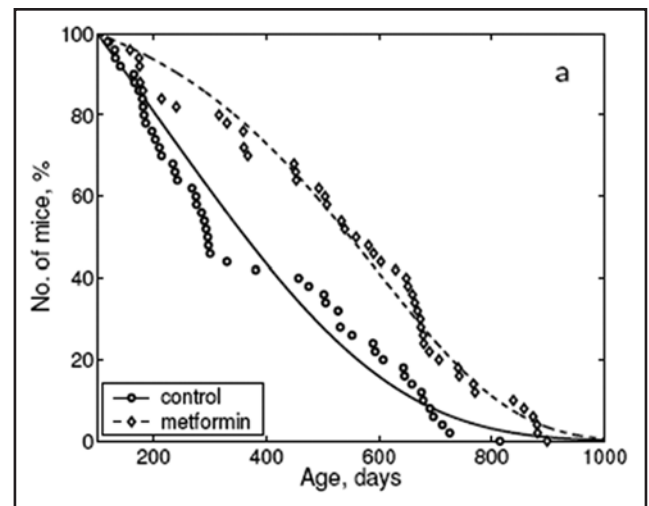


Figure 3. Survival curves of female SHR mice treated and not treated with metformin.

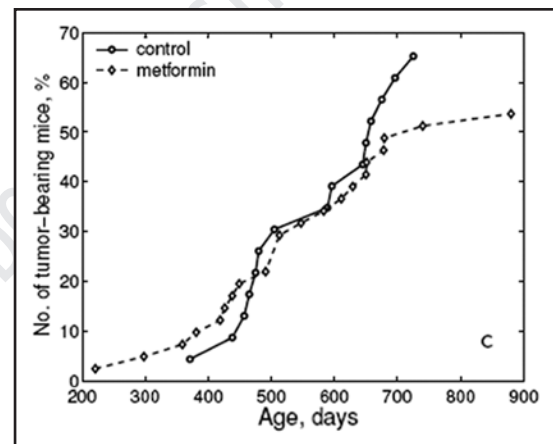


Figure 4. Tumor yield curves of female SHR mice treated and not treated with metformin.

Like other biguanides, metformin slows down the age-related disturbances in the estrous function of mice in our experiments. It is worthy of note that metformin improves menstrual regularity, leading to spontaneous ovulation, and enhances the induction of ovulation with clomiphene citrate in women with polycystic ovary syndrome.^{32,33}

The treatment with metformin only slightly influenced body weight and food consumption of SHR mice as compared with intact control.

The treatment with metformin failed to influence total tumor incidence and latency in the present experiment. In transgenic HER-2/neu mice, the similar treatment with metformin not changed the incidence of mammary adenocarcinomas however increased their latency and decreased the multiplicity.¹³ Antidiabetic biguanides phenformin and buformin inhibited spontaneous and chemically induced carcinogenesis in a number of the models (reviewed in ref. 12). The decrease of breast carcinoma risk has been reported in diabetes mellitus type 2 patients receiving biguanides.³⁴ It is possible that the absence of metformin effect on tumor incidence in the present experiment was incidental. The new experiments with metformin in SHR mice and some other mouse strains will give more certain response to this question.

Table 4 Incidence and latency of tumors in female SHR mice treated and not treated with metformin

Parameters	Control	Metformin
Effective number of mice	23	40
Number of tumor-bearing mice	15 (65%)	22 (55%)
Number of malignant tumor-bearing mice	9 (39.1%)	17 (42.5%)
Total number of tumors	15	26
Mean life span of total tumor bearing mice, days	617 ± 26.4	588 ± 31.0
Mean life span of fatal tumor bearing mice, days	631 ± 37.9	556 ± 30.3

Although it is known that free radicals are produced during metabolic reactions, it is largely unknown which factor(s), of physiological or pathophysiological significance, modulate their production in vivo. It has been suggested (though this conclusion is not unanimous) that hyperinsulinemia may increase free radicals and therefore promote aging, independent of glycemia.^{2,4,35} Plasma levels of lipid hydroperoxides are higher, and antioxidant vitamins are lower in individuals who are resistant to insulin-stimulated glucose disposal but otherwise glucose tolerant, nonobese, and normotensive.³⁵ This finding indicates that enhanced oxidative stress is present before diabetes ensues and therefore cannot simply be explained by overt hyperglycemia. There is substantial evidence supporting the hypothesis that selective resistance to insulin-stimulated (muscle) glucose disposal and the consequential compensatory hyperinsulinemia trigger a variety of metabolic effects, likely resulting in accelerated oxidative stress and aging.^{2,4}

The anti-diabetics biguanides inhibit fatty acid oxidation, suppress gluconeogenesis in the liver, increase the availability of insulin receptors, inhibit monoamine oxidase,³⁶ increase sensitivity of hypothalamo-pituitary complex to negative feedback inhibition, reduce excretion of glucocorticoid metabolites.⁴ It was shown that metformin decreases platelet superoxide anion production in diabetic patients.³⁷

It is worthy to note, that experiments in yeast and *C.elegans* show that the life extension by CR is not a mechanical output of low calories and consequence of a reduction in ROS or AGE formation, but a process that is highly regulated, triggering metabolic shift toward respiration that activates the regulator SIR2.³⁸ In yeast and worms, life span is extended by extracopies of *SIR2/Sir-2.1* gene,³⁹ by *SIR2* orthologue, *Sirt1* (sirtuin 1),⁴⁰ or by small molecule sirtuin-1 agonists, e.g., resveratrol.⁴¹ In mammals, it is suggested that SIRT1 is a key regulator of cell defence and survival in response to stress.^{42,43} Recently it was shown that the expression of mammalian Sir2 (SIRT1) is induced in CR rats as well as in human cells that were treated with serum from these animals.⁴⁴ Long-lived mutant mice and CR rodents are protected from cancer despite attenuating apoptosis possibly because their cells possess increased defence and repair mechanisms and they retain the ability to undergo apoptosis if the damage is beyond repair.⁴⁴ It was observed that phenformin inhibited proliferation and induced enhanced and transient expression of the cell cycle inhibitor p21 and apoptosis in human tumor cells lines.⁴⁵

Therefore, at least some antidiabetic drugs could be potent geroprotectors and anticarcinogens. Research in this direction warrants further experimental as well as clinical^{16,46} efforts.

Material and methods

Animals. Outbred Swiss-derived female SHR mice were purchased from the "Rappolovo" Animal Farm of the Russian Academy of Medical Sciences. The mice were kept in groups of 5–7 animals in polypropylene cages (30 x 21 x 10 cm) under standard light/dark regimen (12 hours light:12 hours darkness) at 22 ± 2°C, and received standard laboratory chow²³ and tap water *ad libitum*.

Experimental design. One hundred and sixty female SHR mice at the age of 3 months were randomly divided into two groups. Mice of the first group were given metformin (Siophor, Berlin-Chemie, Menarini Group) with drinking water (100 mg/kg) daily, whereas the mice of the second group were given tap water without metformin and served as a control. This dose of metformin is similar to used in our earlier experiments with HER-2/neu mice²⁰ and equal to 300 mg/m² of the surface area. Recalculation for humans gives in average 510 mg/m², that is much less than commonly used in clinical practice (1.0–2.5 g per day). Once a week all mice were palpated for detection of mammary tumors appearance. The localization and the size of tumors were registered on the special charts. Once a month all mice were weighted and, simultaneously, the amount of consumed food and water was measured, and the rates of the consumed water (ml) and food (g) per mouse and per body weight unit were calculated. Once in every 3 months, daily for 2 weeks vaginal smears of the animals were examined cytologically to estimate the estrous function. In the same period, rectal body temperatures of the mice were measured with an electronic thermometer, TPEM (KMIZ, Russia).

The time of appearance of mammary tumors was evaluated by palpation and the neoplastic masses were measured with calipers in the two perpendicular diameters. Progressively growing masses of >3 mm in mean diameter were regarded as tumors. At the age of 3, 9 and 12 months 10 mice from each group were sacrificed by decapitation after overnight starvation. Samples of serum were obtained and stored at the -20°C for subsequent analyses. Other animals were observed until their natural deaths. The date of each death was registered, and the mean life span, the age at which 90% of the animals died, and the maximum life span were estimated.

Metabolic and hormonal assays. The serum levels of glucose were estimated by enzymocolorimetric (glucose-oxidase) method with kits from "Impact" (Moscow, Russia); cholesterol and triglycerides—by enzymocolorimetric method with kits of "Olvex" (St.Petersburg, Russia); insulin—by immune enzyme assay (ELISA) with kits from Diagnostic Systems Laboratories, Inc., (U.S.A.); estradiol-17β by ELISA with kit from DRG (Germany).

Pathomorphological examination. All animals were autopsied. Site, number and size of mammary tumors and their metastases in lungs were checked. All tumors, as well as the tissues and organs with suspected tumor development were excised and fixed in 10% neutral formalin. After the routine histological processing the tissues were embedded into paraffin. 5–7 μm thin histological sections were stained with haematoxylin and eosine and examined microscopically. Tumors were classified according to International Agency for Research on Cancer recommendations.²⁴

Statistics. Experimental results were statistically processed by the methods of variation statistics with the use of STATGRAPH statistic program kit. The significance of the discrepancies was defined according to the Student t-criterion, Fischer exact method, χ², non-parametric Wilcoxon-Mann-Whitney and Friedman RM Anova on

Ranks. Student-Newman-Keuls Method was used for all pairwise multiple comparisons. Coefficient of correlation was estimated by Spearman method.²⁵ Differences in tumor incidence were evaluated by the Mantel-Haenszel log-rank test.

Parameters of Gompertz model were estimated using maximum likelihood method, non-linear optimization procedure²⁶ and self-written code in 'Matlab'; confidence intervals for the parameters were obtained using the bootstrap method.²⁷

For experimental group Cox regression model²⁸ was used to estimate relative risk of death and tumor development under the treatment compared to the control group: $h(t, z) = h_0(t) \exp(z\beta)$, where $h(t, z)$ and $h_0(t)$ denote the conditional hazard and baseline hazard rates, respectively, β is the unknown parameter for treatment group, and z takes values 0 and 1, being an indicator variable for two samples—the control and treatment group.

Semiparametric model of heterogeneous mortality²⁹ was used to estimate the influence of the treatment on frailty distribution and baseline hazard.

Acknowledgements

This article was supported in part by grants # 05-04-48110 and 08-04-00116 from the Russian Foundation for Basic Research, and by grant # NSh-5054.2006.4 from the President of Russian Federation.

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